CENGAGE | WEBASSIGN

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FSK116 S1 2021, section 1, Spring 2021

Online Test 2 (Test)

Current Score

QUESTION | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10

Instructions

Answer all questions. You have two submission opportunities
You have 100 minutes to do the test.

Description

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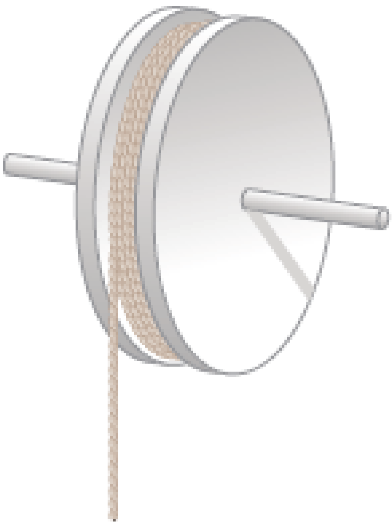
1. DETAILS **SERPSE10 10.CQ.004.** MY NOTE

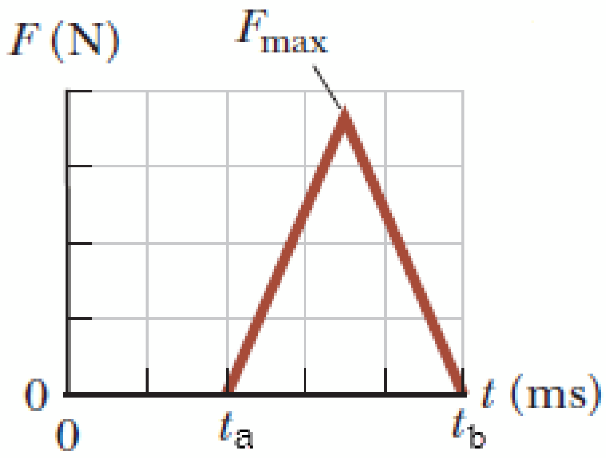
Explain how you might use the apparatus described in the figure below to d
the wheel. *Note:* If the wheel does not have a uniform mass density, the m
equal to $\frac{1}{2}MR^2$.



The moment of inertia can be found by pulling on the string to see how eas
difficult the wheel rotates. The further the mass is away from the point of
rotation, the bigger the moment of inertia according to the equation. This i
therefore an acurate way of measuring as the mass is close to the point of
rotation.

2. DETAILS **SERPSE10 9.3.OP.006.** MY NOTE

An estimated force–time curve for a baseball struck by a bat is shown in the
F_{max} = 20,000 N, t_a = 1.5 ms, and t_b = 3 ms. From this curve, determine t



(a) the magnitude of the impulse delivered to the ball

15 N · s

(b) the average force exerted on the ball

10 kN

3. DETAILS **SERPSE10 9.3.P.009.MI.SA.** MY NOTE

This question has several parts that must be completed sequentially. If you will not receive any points for the skipped part, and you will not be able to

Tutorial Exercise

The front 1.20 m of a 1,201-kg car is designed as a "crumple zone" that col collision.

(a) If a car traveling 24.0 m/s stops uniformly in 1.20 m, how long

(b) What is the magnitude of the average force on the car?

(c) What is the magnitude of the acceleration of the car? Express th acceleration of gravity.

Step 1

(a) If we assume a uniform deceleration of the car, then the average speed distance the car travels as it stops uniformly and collapses to absorb the sh the collision Δt is the time required for the car to travel the 1.20-m length c given by

$$\Delta t = \frac{\Delta x}{v_{av}} = \frac{\Delta x}{(v_f + v_i)/2} = \frac{2\Delta x}{v_f + v_i}$$

$$= \frac{2\left(\boxed{1.20} \quad \boxed{1.2}\ \text{m}\right)}{\left(0 + \boxed{24.0} \quad \boxed{24}\ \text{m/s}\right)} = \boxed{10} \quad \boxed{10} \times 10^{-2}$$

Step 2

(b) From the impulse-momentum theorem, $\vec{\mathbf{I}} = \vec{\mathbf{F}}_{av}\Delta t = \Delta\vec{\mathbf{P}}$, we know that the change in momentum to the corresponding time interval. The magnitud the car during the impact is

$$F_{av} = \frac{\left|\Delta\vec{\mathbf{p}}\right|}{\Delta t} = \frac{m\left|\vec{\mathbf{v}}_f - \vec{\mathbf{v}}_i\right|}{\Delta t}$$

$$= \frac{\left(\boxed{1201}\ \checkmark\ \boxed{1200}\ \text{kg}\right)\left|0 - \boxed{24.0}\ \checkmark\ \boxed{24}\ \text{m/s}\right.}{\left(\boxed{10}\ \checkmark\ \boxed{10}\ \times 10^{-2}\ \text{s}\right)}$$

Step 3

(c) The magnitude of the average acceleration a_{av} is the ratio of the change
the collision.

$$a_{av} = \frac{\left|\vec{\mathbf{v}}_f - \vec{\mathbf{v}}_i\right|}{\Delta t} = \frac{\left|0 - \boxed{24.0}\ \checkmark\ \boxed{24}\ \text{m/s}\right|}{\left(\boxed{10}\ \checkmark\ \boxed{10}\ \times 10^{-2}\ \text{s}\right)} = \boxed{240}\ \checkmark$$

Step 4

We are asked to express the acceleration as a multiple of the acceleration o
acceleration in terms of g. We have

$$a_g = a_{av}\left(\frac{1g}{9.80\ \text{m/s}^2}\right)$$

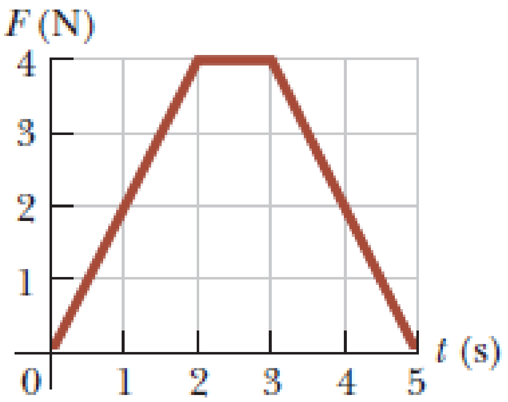
$$= \left(\boxed{240}\ \checkmark\ \boxed{240}\ \text{m/s}^2\right)\left(\frac{1g}{9.80\ \text{m/s}^2}\right) = \boxed{24.5}\ \checkmark$$

SUMMARY

Since we were given the "crumple zone" distance of the car, we were able t
collision from the average speed, assuming constant deceleration. We used
to calculate the magnitude of the average force on the car during impact. W
acceleration from the ratio of the change in the car's speed to the duration
converted the average acceleration to a multiple of g, the acceleration due
of the collision is key to reducing the average force on the passenger involv

4. DETAILS **SERPSE10 9.3.P.010.** MY NOTE

The magnitude of the net force exerted in the *x* direction on a 2.70-kg parti
figure below.



(a) Find the impulse of the force over the 5.00-s time interval.

$\vec{I}$ =

$$12.0ˆi

N · s

(b) Find the final velocity the particle attains if it is originally at rest

$\vec{v}_f$ =

4.44ˆi

m/s

(c) Find its final velocity if its original velocity is -2.40 $\hat{i}$ m/s.

$\vec{v}_f$ =

2.04ˆi

m/s

(d) Find the average force exerted on the particle for the time inter

$\vec{\mathbf{F}}_{avg}$ =

$\hat{i}$ 2.40 $\hat{i}$

N

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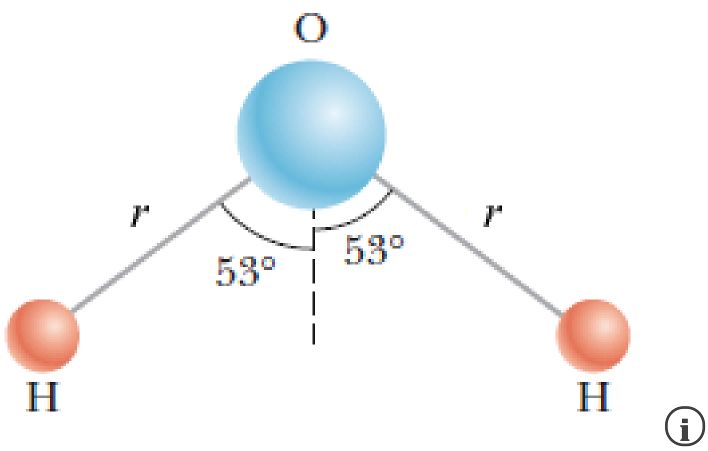
5. DETAILS **SERPSE10 9.6.OP.026.** MY NOTE

Consider the schematic of the molecule shown, with two hydrogen atoms, H,
angle between the two bonds is 106°. If the bond length r = 0.104 nm long,
molecule. The mass m_H of the hydrogen atom is 1.008 u, and the mass m_O o
(Use a coordinate system centered in the oxygen atom, with the x-axis to the
the coordinates of the center of mass in nm.)

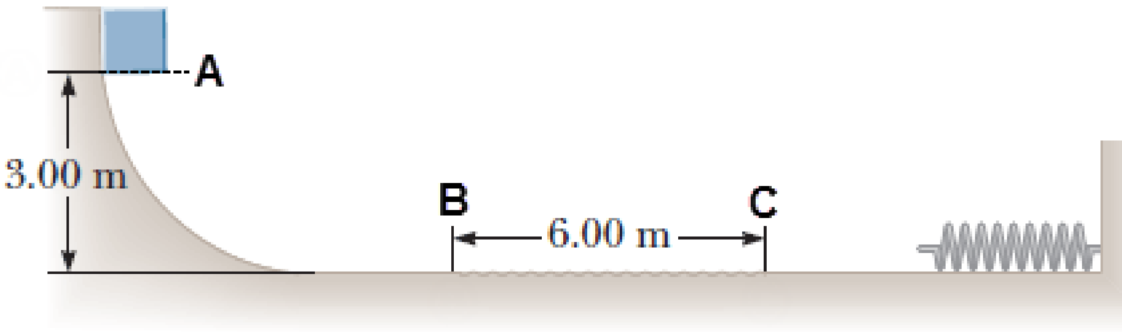


x_{CM} = 0 nm

y_{CM} = 0.007 nm

6. DETAILS **SERPSE10 8.A.OP.031.MI.** MY NOTE

(a) A 15.0 kg block is released from rest at point A in the figure below. The
portion between points B and C, which has a length of 6.00 m. The bloc
spring of force constant 2,200 N/m, and compresses the spring 0.350 r
before coming to rest momentarily. Determine the coefficient of kinetic
rough surface between points B and C.



(b) **What If?** The spring now expands, forcing the block back to the left. D

- Yes
- No

If the block does reach point B, how far up the curved portion of the tra
how far short of point B does the block come to a stop? (Enter your ans

m

7. DETAILS SERPSE10 10.2.OP.004.MI. MY NOTE

An electric motor rotating a workshop grinding wheel at 1.20×10^2 rev/mir has a constant negative angular acceleration of magnitude 2.04 rad/s^2.

(a) How long does it take the grinding wheel to stop?

6.16 s

(b) Through how many radians has the wheel turned during the tim

38.70 rad

8. DETAILS SERPSE10 15.3.OP.013. MY NOTE

A plastic ball attached to a spring moves in simple harmonic motion. The amp 10.0 cm, and the spring constant is 5.00 N/m. When the ball is halfway betw maximum displacement from equilibrium, its speed is 17.7 cm/s.

(a) What is the mass of the ball (in kg)?

1.20 kg

(b) What is the period of oscillation (in s)?

0.10 s

(c) What is the maximum acceleration of the ball? (Enter the magnitude in

-0.21 m/s^2

9. DETAILS **SERPSE10 15.2.OP.002.MI.** MY NOTE

A vertical spring stretches 3.4 cm when a 6-g object is hung from it. The ob
mass 26 g that oscillates up and down in simple harmonic motion. Calculate

[ ] s

10. DETAILS **SERPSE10 15.2.OP.003.** MY NOTE

In an engine, a piston oscillates with simple harmonic motion so that its po
expression,

$$x = 8.00 \cos\left(t + \frac{\pi}{8}\right)$$

where x is in centimeters and t is in seconds.

(a) At $t = 0$, find the position of the piston.

8.00 ✗ cm

(b) At $t = 0$, find velocity of the piston.

0.05 ✗ cm/s

(c) At $t = 0$, find acceleration of the piston.

-8.00 ✗ cm/s^2

(d) Find the period and amplitude of the motion.

period [ ] s

amplitude 8.00 ✓ cm

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